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Deep Learning Approach for Forecasting Athletes' Performance in Sports Tournaments

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Deep Learning Approach for Forecasting Athletes' Performance in Sports Tournaments

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ABSTRACT

Sports and international tournaments have gained world attention in the past decade. Enhancing sports activities and promoting sports to participate in international events, competitions, and tournaments play a substantial role in the development and advancement of nations around the globe. In this paper, we applied different deep learning models for predicting athletes' performance in tournaments to help them improve their results. We propose a deep learning selection algorithm to evaluate the effectiveness of the athletes' current training by predicting their race results upon completing each additional training, which potentially improves their performance. We gathered public training data for athletes who participated in the 2017 Boston Marathon within a five-month window prior to the race. Deep learning models were applied and evaluated to predict marathon finishing times. These include Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Results show that Deep Learning models give improved race time prediction accuracy over the baseline machine learning model, such as standard Linear Regression (LR).

CCS CONCEPTS

•Computing methodologies → Machine learning → Machine learning approaches → Neural networks

KEYWORDS

Running, sports training, prediction modeling, deep learning, marathon

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1 Introduction

Over the past few years, marathon running sport has attracted athletes around the world. The number of People finishing full long-distance races has expanded by more than 40% this decade [13]. Therefore, it is important to prepare the athletes to achieve their best in such national and international tournaments. Workout and practice are the major factors that affect the athlete's performance and thus they need to be correctly applied. However, there is a lack of studies and follow up research in evaluating the performance of sports participants before going into competitions.

Several factors can influence the athlete's performance in a competition, among which are cramps, starting pace too fast with respect to practice preparation, dehydration, and others. Accurately estimating the athlete's performance will help them improve their finishing times during races. The lack of attention to the running analytics literature encouraged us to research the prospective applications of data analytics in the field of endurance running.

The main research questions to be investigated in this paper are:

- Will deep learning models give better accuracy in the prediction of athletes' performance in competitions?
- What parameters and metrics affect sports performance?
- What are workout activities better for performance?

The main objectives of this research can be summarized into: 1) Maximize the winning chances in national and international sports competitions, and predict the athletes' performance in tournaments and major competitions. 2) Apply and compare different Deep Learning techniques to predict the athletes' performance in tournaments. 3) Test the algorithms on a large dataset and evaluate the applicability of our solution in a real environment.

This paper is organized as follows: the following two sections depict the background summary of the current state of knowledge and review the related work. After that, in Section 4 we describe our proposed model to predict the athletes' finish time in marathons based on the evaluation of multiple deep learning prediction models. Section 5 provides the experimentation and model evaluation. Finally, the last section concludes this paper and recommends future directions.

2 Background on Deep Learning Models

In this paper, we propose a model for predicting athletes' performance in preparation for major competitions. Accurately predicting athletes' performance for tournaments affects not only their performance but also their practices and lifestyle. Through our model, we provide athletes with a better prediction for their performance.

In this research, we investigate and assess different Deep Learning models and we select the most appropriate approach that provides better accuracy for predicting athletes' performance in competitions. The different Deep Learning models we are considering in this paper include Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU).

RNN is a class of artificial neural networks that supports sequential data, such as time series. Compared to a standard feedforward neural network, RNN takes as inputs not only data at the current time step but also the output of the previous time step [2]. Therefore, it keeps a memory of the previous state and uses it as an input to the network to influence predictions at the current time step. However, standard RNNs express some difficulty with long-term dependencies [10].

LSTMs is a modified version of RNN that overcomes the problem of long-term dependencies by preserving the error that can be back-propagated through time and via layers. This allows the support of model training over long sequences and makes it best suited for prediction, classification, and generation of accurate sequence data [15].

GRU is a variation of the LSTM that was introduced in 2014 by Kyunghyun Cho et al. [3]. Its learning phase requires a lower number of parameters, computations to make updates to its hidden state, and are faster to train [4].

3 Related Work

In this section, we survey different approaches for athletes' performance prediction proposed in the literature. There are a few research initiatives in the area of athletes' race time prediction based on historical practice information and physical characteristics of the athletes.

Authors in [24] used Riegel formula to predict race time for races up to a half-marathon with good results. However, their results show 10 min less than the actual race time for 50% of the data which was collected using an Internet survey. In this work, sex, age, BMI, race training, and mean race velocity were used for prediction. Other research work used the data for twenty-two runners who participated in an overall number of 46 races and an 8-week training period. This data was used to predict the mean marathon pace using a correlation mathematical approach [21]. The results show that the mean distance per week and the mean training pace are strongly correlated with the mean pace time. Another work proposed in [19] used multi-variate analysis on athlete characteristics as well as training data to predict the race time. The athletes' characteristics include body mass, body height, body mass index, circumferences of limbs, thicknesses of skin-folds and body fat. On the other hand, the investigated training information is composed of volume and speed in the training. The study was intended to find the highly correlated predictors for race time of female athletes. The results show that both low calf circumference and high running speed during training are highly associated with better race time. The authors in [6] proposed a model to predict Olympic athletes' sprint time for 100 m races. This model used the Olympic official records of the world's best performances from the year 1896 to the year 2016. The authors adopted a polynomial linear model of degree 7 for predicting the race time in this system.

Famous international marathons such as Boston and London marathons are good sources for athletes' data in research work related to athletes' performance. For example, the authors in [17] used the 2015 Boston and London marathons runners' data. This data includes distance, time, net elevation gain and pace for

TABLE I. SUMMARY OF REVIEWED LITERATURE THAT PROPOSES MARATHON RACE TIME PREDICTION FOR ATHLETES

Paper Title	Year Published	Race Type	Prediction	Number of Participants	List of Variables Used
[8]	2017	Half Marathon	Performance (Race Time)	78	VO2Max, Speed At The Anaerobic Threshold, Peak Speed, Contact And Flight Times, Step Length, Step Rate
[7]	2019	Full Marathon	Performance (Race Time)	16	Aerobic threshold (AeT), anaerobic threshold (AnT), maximal oxygen uptake (VO2 Max), running economy
[11]	2014	Half Marathon	Race Time	230	Running speed, midaxillary skinfold, body mass index, percent body fat
[12]	2011	Ultra Marathon	Distance Run in 24 hour	63	Longest Training Session, km, Personal Best Marathon Time, Minutes
[16]	1984	Full Marathon	Race Time	18	Running velocity at the anaerobic threshold
[25]	2013	Full & Half	No Prediction (only Statistical analysis)	147 & 126	Runner characteristics: Age, Body mass, height, BMI, circumference of calf and arm...etc. Training characteristics: Years as an active runner, running hours...etc.
[18]	2011	Half Marathon	Race Time	84	Runner characteristics: Age, Body mass, height, BMI, circumference of calf and arm...etc. Training characteristics: Years as an active runner, weekly distance ran...etc.

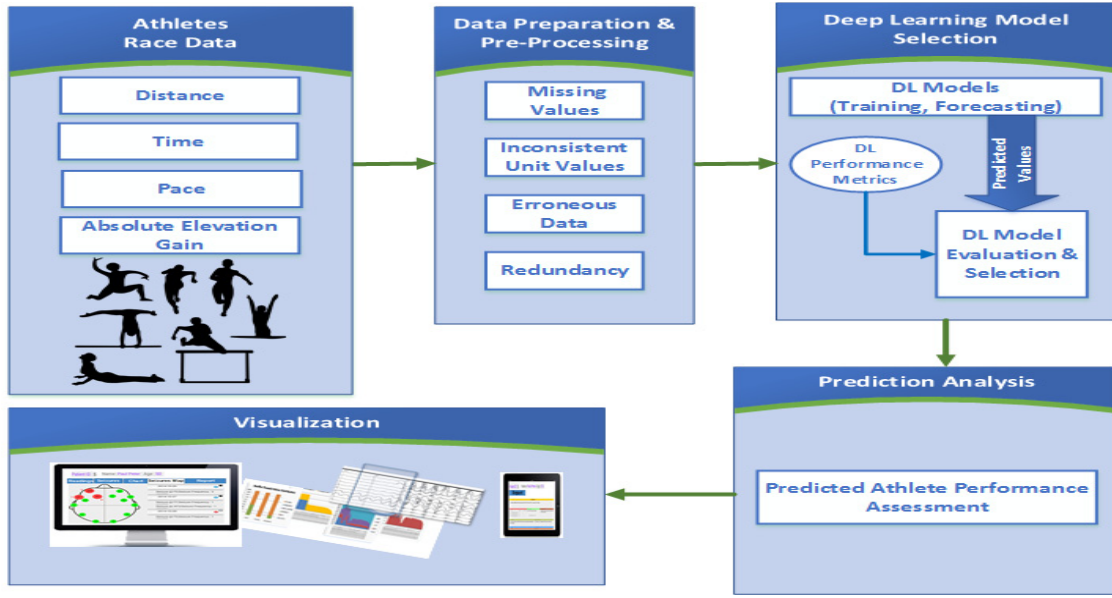


Figure 1: Deep Learning Model Selection for Athletes' Performance Prediction

different workouts. Results proved that Bagging algorithms give good race time prediction for amateur athletes compared to zeroR

Algorithm 1 Deep Learning Prediction-Based Athlete Performance Algorithm

```

1: Input:
   athleteList: List of Athletes including their workout sessions,
   Each workout has activity details List per Workout (e.g. distance, moving/elapsed time, pace, elevation gain, and calories burned.
   predictionModels: List of DL Prediction Models
2: Output:
   PredictedT: Race Time Prediction,
   selectedModel: Selected DL Model
3: procedure ATHLETERACETIMEMULTIMODELPREDICTION(athleteList,
   predictionModels)
4:   minMAPE  $\leftarrow \infty$ 
5:   for all  $m \in \text{predictionModels}$  do
6:     for all  $a \in \text{athleteList}$  do
7:        $W \leftarrow \text{getWorkoutList}(a)$ 
8:       for all  $w \in W$  do
9:          $\hat{t} \leftarrow \text{Predict}(t, m)$ 
10:         $\text{MAPE} \leftarrow \text{evalMAPE}(\hat{t}, w)$ 
11:       end for
12:     end for
13:     if ( $\text{MAPE} < \text{minMAPE}$ )
14:        $\text{minMAPE} \leftarrow \text{MAPE}$ 
15:        $\text{PredictedT} \leftarrow \hat{t}$ 
16:        $\text{selectedModel} \leftarrow m$ 
17:     endif
18:   end for
19:   return PredictedT, selectedModel
20: end procedure

```

and linear regression algorithms. Additionally, the authors in [1] used athletes' data collected from Strava which includes runner's pace, heart rate (HR) and cadence during the marathon for predicting the pace race. They adopted an automated time series feature extraction method called tsfresh. They provide personalized strategy race guidance for athletes.

TABLE I. summaries some extra works that tackled the marathon race time prediction for athletes. The table highlights the size of the dataset represented by the number of participants, the list of variables and attributes used for prediction, type of marathon, and target variable to be predicted. All proposals show a fairly low number of participants which affects the accuracy of the proposed models. The majority of these proposals use vital signs measured variables (e.g. heart rate, Oxygen saturation...etc.), which are impractical to measure during each workout.

Although the above research approaches adopt different forecasting algorithms for race prediction, very few adopt machine learning for race prediction. Authors in [23] adopted a Case-based reasoning (CBR) technique for race time prediction. However, none of the aforementioned surveyed literature has used Deep Learning models for race prediction, the latter has the potential to give higher accuracy predictions for race times while handling a huge dataset. Accordingly, extra efforts are needed to experiment with other models such as deep learning algorithms to achieve higher accuracy and provide athletes with a better prediction for their marathon finish time.

4 Deep Learning Model Selection for Athletes' Performance Prediction

In this section, we describe our proposed deep learning model selection algorithm that uses athletes' workout data to predict their performance in tournaments. Our goal is to help athletes to improve their performance by predicting their race time in an upcoming marathon. Predicting the race times will act as a motivation and a drive for the athlete to persist in and improve their activities to achieve their target results.

We propose an approach that collects athletes' data, applies data preprocessing, selects the most accurate Deep Learning model, and

uses the selected model to predict athletes' performance in a competition. Figure 1 depicts our proposed solution and its key modules. The key two components are the Multi-Model Selection module and the Prediction Analysis module. These will be detailed in the following subsections.

4.1 Model Description

In this model, we first determine what workout activities and select the training metrics, which best improve athlete performance. Accordingly, we select training runs metrics like distance, time length, pace, and elevation gain.

Following, data collection and data preprocessing, we apply different prediction models and Deep Learning techniques on preprocessed athletes data, such as RNN, GRU, and LSTM, to achieve the most accurate and interpretable model for predicting athlete performance in a marathon. This process is performed in the deep learning model selection component. In this component, we evaluate different Deep Learning techniques for predicting race time. This is performed in the DL models module. Then we select the best model that gives the highest prediction accuracy. This process is the responsibility of the DL Model evaluation and selection module. The selection process is detailed in the following subsection. The second key component is the prediction analysis resulted from the analysis of predicted performance.

4.2 DL Selection Evaluation

In this section, we describe our Model selection algorithm for predicting marathon-finishing times. After applying different DL models to the existing data, the results predictions are evaluated against testing datasets. In this work, we use the following models: Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). We perform automatic selection of one of the aforementioned Deep Learning prediction models. These chosen models fit the timely recorded athletes' performance during workouts, represented as time series variables. Thus, the data is fed into different Deep learning models for predicting future finish time values. Then, the algorithm evaluates the prediction results against several DL performance metrics, such as Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), and Mean absolute percentage error (MAPE), to select the model, which has the highest prediction accuracy. The following formulas define the aforementioned measurements:

$$RMSE = \sqrt{\frac{\sum_{t=1}^n (y_{pt} - y_t)^2}{n}}, \quad (1)$$

$$MAE = \sum_{t=1}^n \frac{|y_{pt} - y_t|}{n}, \quad (2)$$

$$MAPE = \frac{100}{n} \sum_{t=1}^n \frac{|y_{pt} - y_t|}{y_t}, \quad (3)$$

where y_{pt} is the predicted value and y_t is the actual value and n is the number of data entries. We measured the MAPE for RNN, LSTM, and GRU used for predicting the marathon finish time to select the prediction algorithm with the highest accuracy for the provided athletes' dataset.

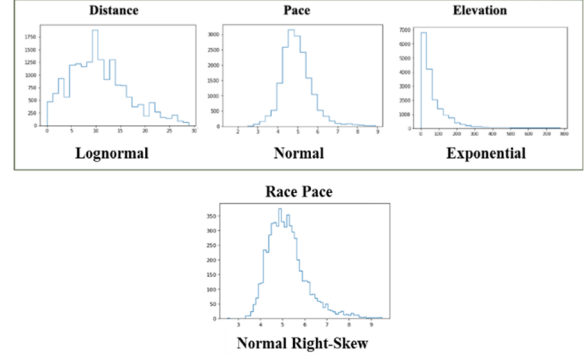


Figure 2: Activity data and race pace distributions

Once the DL model is selected, its results are assessed against optimum race finish time scores. This algorithm is depicted in Algorithm 1. The input of the algorithm is the athletes' dataset of workouts, and the candidate DL models. The output is the predicted race finish time. The algorithm compares the predicted finish time generated by different DL models, based on MAPE to select the model giving the best accuracy.

5 Experiments and Evaluation

We describe in this section, the experimental evaluation we conducted to assess our prediction models. Therefore, we first describe the experimentation setup, then the dataset collection process, and the data preparation and pre-processing. Finally, we provide analysis and discussion of the results.

5.1 Experimentation Setup

We have implemented the three Deep Learning models using PyTorch; an end-to-end open source framework for machine learning [5][14]. Moreover, we use Jupyter Notebook, which is another open-source web application that enables users to perform different tasks, such as statistical modeling, data visualization, and machine learning. Our baseline model for comparison is the standard linear regression machine learning.

5.2 Dataset

In this paper, we choose marathon-running data to test our proposed model. For an average person, running is considered the easiest sport. This is because it is suitable for a busy lifestyle and can be practiced anywhere and anytime. Therefore, endurance running has become a popular sport internationally.

Our data includes training sessions from athletes who participated in the 2017 Boston Marathon during their five-month period of training for our experiments. The Boston Marathon is one of the most historic and renowned marathons in the world with a large number of participants. Therefore, using the Boston Marathon participants' data will provide a large-sized dataset, which when used will help to produce more accurate results.

The data collected contains Athlete name, date of workout, running distance in meters, time of running duration, pace in minute per km, average pace of the athlete in this workout, and the elevation, which

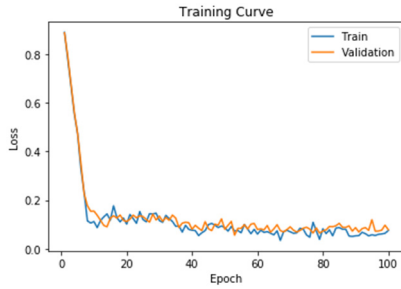


Figure 3: RNN-HS50 Prediction Error

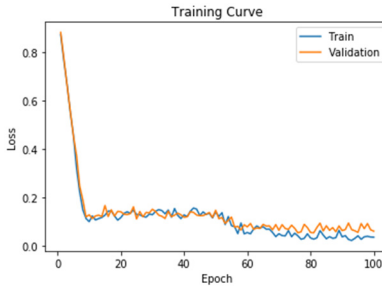


Figure 4: GRU-HS50 Prediction Error

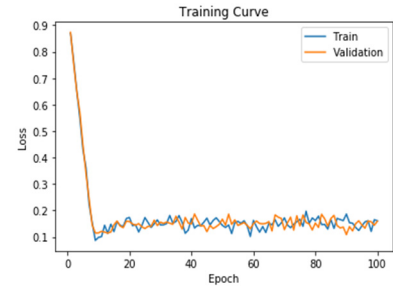


Figure 5: LSTM-HS50 Prediction Error

is the total of all elevation increments throughout the workout. We collected 549 athletes who ran the Boston Marathon in April 2017 and 23,763 training activities performed 5 months prior to the race.

Figure 2 depicts the activity data distributions and race pace distribution. As shown in the figure, the distance, pace, and elevation are lognormal, normal, and exponential, respectively. Furthermore, the marathon race pace was also normally distributed, but right-skewed to some extent.

5.3 Data Preparation and Pre-Processing

The data we used needs preparation and pre-processing to be fed into the ML models. We used Talend data preparation tool [20] by importing the athletes' information files through its interface. It supports functionalities like parsing files and providing few suggestions for data cleansing, such as tips to replace missing data as well as enrich it. It is a visual tool with graphical charts that also measures some data quality metrics.

Additionally, Trifacta Wrangler (version 3.0.1-client1+push2) is used for data preparation, pre-processing and Data Quality metrics measurements [22]. Furthermore, we used different libraries for data pre-processing, such as datetime for time-related data, and pandas for data cleansing and normalization.

5.4 Data Analysis and Result Discussion

In this section, we evaluate our proposed model by testing the accuracy of each of our proposed prediction algorithms to make sure that our model selection is supported by the most accurate prediction model.

TABLE II. DIFFERENT DEEP LEARNING ALGORITHMS EVALUATION COMPARED TO BASELINE (LR)

Performance Quality Metric		MAPE			
Model Description		LR	RNN	GRU	LSTM
Without Outliers	HS 25	0.1	0.07	0.06	0.1
	HS 50		0.056	0.052	0.11
Full Dataset	HS 25	0.11	0.11	0.089	0.122
	HS 50		0.12	0.1	0.104

We have implemented RNN, LSTM, and GRU using 25 hidden states and compared the results to the same model having 50 hidden

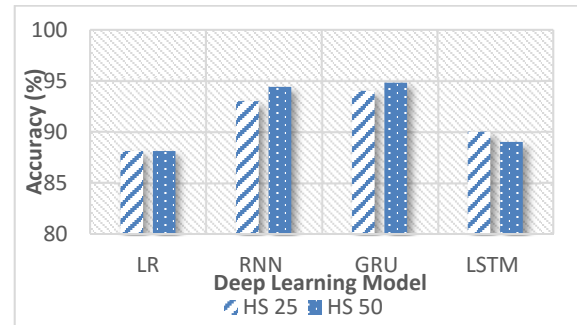


Figure 6: Prediction Accuracy

states. The models adopting 50 hidden states showed slight accuracy improvement. However, we decided to apply these DL models on the data after removing the outliers using the statistical quality control three-sigma rule [9]. This is achieved by eliminating any values that were above the mean by 3σ and below the mean by 3σ (i.e. $\pm 3\sigma$, yielding 93.3% of the dataset). Removing the outliers depicted a significant improvement of the results. This is illustrated in TABLE II. In this table, we recorded the MAPE values for each tested model.

For our dataset, the DL selection shows that GRU with 50 hidden states gives the least MAPE value i.e. performs better than the other algorithms. Therefore, the selection module chooses it for predicting the race finish time.

Figures 3, 4, and 5 depict the training curve of prediction error (MAPE) for RNN, GRU, and LSTM with 50 hidden states trained on the dataset with no outliers respectively.

Figure 3 shows that the RNN model quickly (15 epochs) learns the best features to predict accurate marathon times, but gets stuck in a local minimum for around 25 epochs. Right before epoch 40, the model finds its way out of the local minimum and gains around 10% accuracy and stabilizes at 0.10 MAPE. Similar to the RNN model, the GRU's training quickly brings the loss down to around 0.18 MAPE and gets stuck in a local minimum. The model exhibits severe symptoms of overfitting after epoch 55. The training loss decreases while the validation curve showed no signs of improvement, which is shown in Figure 4. However, in Figure 5, the LSTM model sharply decreases to a very small loss value, but slowly loses accuracy as it trains more and more. It stabilizes around 0.15 MAPE and shows no signs of improvement.

We also compared the accuracy among competing deep learning models based on the MAPE evaluation method. As shown in Figure 6, 50 hidden state GRU model exhibits the best performance at 95% accuracy versus RNN and LSTM at 94% and 89% respectively after removing the outliers from the dataset. Moreover, it is clearly shown that all DL models outperform the baseline standard LR model.

6 Conclusion and Future Work

Promoting the performance of athletes in national and international competitions contributes to the development of a cohesive society and preserved identity worldwide. This can be achieved by studying their current performance, analyzing practice workouts, and predict their performance in future competitions in order to allow for the improvement of sports achievements. Such motion can be achieved through systematic study and modeling which aims to suggest workout and training activities to improve the athlete's standards and eventually their performance.

This paper provides an insight into the usage of various Deep Learning models to accurately predict athlete performance in competitions and tournaments, which will act as an extrinsic motivation for them to carry on with and improve their training activities to achieve their desired race times. Hence, increase their winning chances. Clarifying this awareness to athletes may affect their training style, allowing them to improve their weaknesses. It is as well, valuable for sports monitoring and achievement through training and workout guidance. It supports athletes through their practice period and incrementally improve their performance in competitions.

The following is a summary of the outcomes of this research: 1) Investigating different Deep Learning models that better predict athlete performance. 2) Deep Learning model selection algorithm for accurately predicting athlete performance in competitions.

Our proposed solution can uniquely provide the following features: it allows multi Deep Learning model selection to predict athletes' performance in competitions. The results depict that Deep Learning models show improved race time prediction accuracy. In particular, the GRU results show improved race time prediction accuracy over RNN, LSTM, and standard linear regression models.

One of the limitations we faced in this research is the limited size of the training data. We suggest using a wider range of athletes training data for runners from different international marathons, such as Tokyo, London, and Berlin marathons. For future work, we suggest extending this work to support athletes with personalized training plans, help in injury prevention, and provide recovery advice. It is also worthwhile to investigate the performance of Random Forest Deep Learning ensemble models with more tuning to achieve higher accuracy results.

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